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October 15th 2022

To: BiGTREE Network

Dear Mr/Ms,

This letter intends to express my interest in the 2-year scholarship offered by your program to attend a biodiversity genomics master's degree and project in Norway. I am convinced this is a perfect opportunity to start an academic career in such a fascinating research topic as it is ancient genomics.

Altogether, my experiences during my studies and after graduating allowed me to develop a passion for biodiversity research through computational tools and a genomics perspective. I discovered an area with many exciting questions worth studying and the possibility of integrating interdisciplinary and diverse groups capable of having a positive impact on nature.

During my undergraduate formation in Genetics and Biotechnology at Universidad Nacional Mayor de San Marcos, my involvement in research groups was driven by the main desire to learn rather than following an inherent or exclusive interest in a particular field. As I defined a better idea of what I wanted to do in the future, I also developed strong technical and organizational skills.

My first research experience was related to human genetics while working as an assistant with Prof. Paredes in the Genetics of Metabolic Diseases Lab. I was in charge of analyzing genetic data from family trios to find an association between polymorphisms with diabetes and other phenotypic measurements. Thanks to this experience, I was familiarized with statistic analysis for genetic association studies and learned to use R.

At the same time, I was invited by Prof. Ore to join the Molecular Ecology Lab., where I mastered routinary laboratory techniques and acquired an interest in bioinformatic tools. Although lack of funding was an obstacle for large projects, I was encouraged to propose and carry out ideas using public data, some of which we shared with the scientific community at conferences. This allowed me to improve my communication skills and my ability to lead student groups. Also, helping senior students with their projects involving diverse organisms such as birds, fishes, or mammals, was important for developing my interest in the conservation of biodiversity, rather than in a single particular organism.

Seeing my enthusiasm for conservation biology during Population Genetics classes, my professor introduced me to biologist Luis Castillo, then-president of RANA, a non-governmental organization dedicated to endangered species conservation. I joined him as a volunteer and assistant, proposing outreach strategies and participating in a scientific expedition to the Andes Highlands, where I was in charge of taking non-invasive samples for DNA analysis from the endangered endemic frog species *Telmatobius macrostomus* and *T. brachydactylus*. This initial fieldwork contributed to reinforcing my idea of how

important genetic studies are for biodiversity conservation, not only to describe and understand the history of endangered populations but also to guide actions taken by decision-makers.

During my senior year, I met Prof. Gisella Orjeda, which presented me with the opportunity to start a project on Antarctic lichens. This experience was an enriching and long journey which included conceptualizing the project, applying successfully for funding, processing samples while facing a pandemic, and passing the dissertation defense for graduation with the highest honors. The study aimed to explore the lichen diversity using DNA barcodes in an understudied region such as Antarctica. As I found novel sequences which could be new species, I am preparing a manuscript in collaboration with a taxonomist to publish the results.

Recently, I joined Prof. Orjeda in the Genomics and Bioinformatics for Biodiversity Lab. as a research assistant. I contributed to devising an ongoing project related to the endangered tree species *Prosopis pallida*, as well as writing grant applications, organizing materials, and recruiting students for the project. The project aims to construct an annotated chromosome-level reference genome assembly and use it as a tool to describe their population genetics and understand the causes of their recent death. We are also studying the microbiomes associated with the tree using metabarcoding. My tasks in the laboratory also include sequencing environmental samples from different projects, training students in preparing Nanopore libraries, and analyzing the sequencing data.

I am certain that following a master's degree program in biodiversity genomics is the next step to continuing my academic path. I find the opportunities offered by the host universities in Norway very interesting, particularly at the Norwegian University of Science and Technology (NTNU). Since I learned how ancient DNA can be used to answer exciting questions concerning the evolutionary history of species and could have an implication for the conservation of present biodiversity, I was determined to become a paleogeneticist. I would be delighted to have the opportunity to work on a project related to paleogenomics under the supervision of Prof. Michael Martin. Having in mind a long-term goal of contributing to biodiversity conservation in Peru, I believe that meeting experienced researchers in Europe would be valuable for future collaboration projects.

Given the opportunity, I will dedicate myself wholeheartedly to making the most out of it by meeting or even exceeding your expectations to contribute to the understanding and conservation of biodiversity. As my performance and achievements in previous experiences can attest, I believe that I am capable of being a valuable addition to the program.

Thank you for your attention.

Sincerely,
Renato La Torre